

WHEATON, MN, USA

WMO: 727533

Lat: 45.780N

Lon: 96.545W

Elev: 312

StdP: 97.63

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-23.9	-32.3	0.2	-27.4	-28.0	0.3	-23.4	13.6	-8.9	12.3	-10.2	3.1	280	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.1	22.6	30.1	21.8	28.1	20.7	24.7	29.6	23.5	28.1	22.3	26.9	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	27.9	22.2	17.5	27.1	21.0	16.3	25.9	76.6	29.7	71.7	27.3	66.9	26.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.8	9.3	DB	-29.8	34.9	3.8	1.4	-32.5	35.9	-34.7	36.8	-36.8	37.6	-39.5	38.6	(4)
(5)			WB	-29.9	26.5	3.5	1.6	-32.4	27.7	-34.5	28.6	-36.5	29.5	-39.0	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-11.1	-9.3	-1.9	6.5	13.8	19.4	21.5	20.0	15.8	8.3	-0.2	-7.9	(6)
(7)		DBStd	12.86	7.86	7.34	7.21	5.58	4.93	3.61	3.04	3.11	4.63	5.11	6.46	6.83	(7)
(8)		HDD10.0	2679	655	541	371	131	20	0	0	0	7	93	309	553	(8)
(9)		HDD18.3	4663	913	774	626	357	157	29	8	18	102	312	556	812	(9)
(10)		CDD10.0	1342	0	0	3	24	138	283	357	311	181	41	4	0	(10)
(11)		CDD18.3	284	0	0	0	1	16	61	107	71	27	2	0	0	(11)
(12)		CDH23.3	2776	0	0	1	20	219	599	962	630	303	41	1	0	(12)
(13)		CDH26.7	824	0	0	0	6	71	187	293	168	87	12	0	0	(13)
(14)	Wind	WSAvg	4.2	4.4	4.5	4.7	4.9	4.9	4.0	3.2	3.2	3.7	4.3	4.5	4.4	(14)
(15)	Precipitation	PrecAvg	668	22	17	33	59	75	107	89	68	79	62	26	20	(15)
(16)		PrecMax	825	76	37	99	115	155	221	187	120	191	177	106	57	(16)
(17)		PrecMin	482	1	2	4	7	31	45	28	14	13	10	0	4	(17)
(18)		PrecStd	97	17	10	21	28	32	48	46	27	46	44	29	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	10.0	18.9	26.2	32.3	33.0	33.0	32.7	32.4	27.9	19.1	8.7	(19)
(20)			MCWB	3.8	6.7	12.7	14.9	19.1	22.6	24.8	24.4	21.1	17.3	12.4	4.5	(20)
(21)		2%	DB	3.8	5.3	14.1	22.1	28.6	31.1	31.3	30.2	28.9	22.8	15.9	5.0	(21)
(22)			MCWB	2.0	2.8	9.9	12.1	17.7	21.5	24.0	23.0	19.8	14.3	9.6	2.0	(22)
(23)		5%	DB	2.1	2.8	11.0	18.6	26.0	28.7	29.2	28.0	27.2	19.1	12.1	2.7	(23)
(24)			MCWB	0.6	1.2	7.6	11.2	15.9	20.2	22.6	21.6	18.9	13.1	8.0	0.8	(24)
(25)		10%	DB	-0.2	1.3	7.4	16.2	22.7	27.1	27.9	27.2	24.1	17.1	8.6	1.2	(25)
(26)			MCWB	-1.0	0.0	5.1	9.7	14.5	19.0	21.6	20.7	17.6	11.5	5.4	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.7	14.0	16.0	20.8	24.7	26.7	26.2	23.5	18.7	13.7	4.9	(27)
(28)			MCDWB	6.4	9.1	17.4	23.5	27.7	30.7	30.9	30.7	28.6	26.3	18.1	8.0	(28)
(29)		2%	WB	2.0	3.1	10.6	13.7	19.0	23.1	25.2	24.2	21.7	16.0	10.2	2.6	(29)
(30)			MCDWB	4.0	5.0	14.3	20.1	25.7	28.5	30.2	28.4	27.2	21.3	14.3	4.6	(30)
(31)		5%	WB	0.4	1.5	7.6	11.8	17.5	21.5	24.0	22.8	20.1	13.7	7.8	1.2	(31)
(32)			MCDWB	1.7	2.9	10.1	17.1	23.2	27.1	28.4	27.2	24.9	18.5	11.5	2.6	(32)
(33)		10%	WB	-1.2	0.0	4.9	10.1	15.7	20.0	22.7	21.6	18.8	11.9	5.5	-0.1	(33)
(34)			MCDWB	-0.4	1.3	7.6	15.3	21.6	24.8	26.9	25.8	23.5	16.0	8.4	1.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	9.4	11.2	11.8	10.9	11.1	11.7	12.6	10.8	9.5	8.4	(35)
(36)			MCDBR	11.0	10.9	14.5	17.0	16.5	14.1	12.6	13.2	15.8	16.3	15.0	11.2	(36)
(37)			MCWBR	9.5	9.0	9.8	9.3	8.1	6.7	6.8	7.3	8.0	9.3	9.8	8.9	(37)
(38)		5% WB	MCDWR	10.4	10.4	13.7	14.8	13.7	11.6	11.6	12.1	13.3	14.3	13.8	10.3	(38)
(39)			MCWBR	9.0	8.6	9.6	9.0	7.6	6.7	7.1	7.5	7.8	8.9	9.5	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.272	0.282	0.311	0.348	0.371	0.385	0.385	0.391	0.357	0.322	0.288	0.267	(40)	
(41)		taud	2.390	2.387	2.424	2.374	2.380	2.385	2.374	2.342	2.428	2.498	2.506	2.431	(41)	
(42)		Ebn at Noon	842	905	922	911	896	882	877	856	860	847	823	811	(42)	
(43)		Edh at Noon	71	87	98	113	117	117	117	116	116	98	79	64	62	(43)
(44)	All-Sky Solar Radiation	RadAvg	1.73	2.83	3.82	4.87	5.52	6.20	6.53	5.47	4.19	2.65	1.71	1.26	(44)	
(45)		RadStd	0.18	0.30	0.35	0.37	0.38	0.36	0.26	0.47	0.28	0.35	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page